

Program : Diploma in Tool and Die Engineering	
Course Code : 2111	Course Title: Production Engineering - 1
Semester : 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To introduce the mechanics of metal cutting and factors affecting machinability
- To provide knowledge on tools and tool materials.
- To explain the working principles of various welding processes.
- To explain working principles of various forging techniques.
- To introduce the different types of patterns and castings.
- To impart knowledge about powder metallurgical products and their applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Engineering Drawings		Engineering Graphics	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO 1	Demonstrate the principles of mechanics of metal cutting and tool life	13	Applying
CO 2	Illustrate types of cutting tool materials, importance of tool geometry and use of cutting fluids	10	Applying
CO 3	Demonstrate bench work and fitting, welding and forging	10	Applying
CO 4	Describe the types of patterns, moulds, and powder metallurgy	10	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						3
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the principles of Mechanics of Metal cutting and tool life		
M1.01	Explain the Mechanics of Metal cutting	4	Applying
M1.02	Illustrate metal cutting processes	3	Applying
M1.03	Describe Tool wear	3	Understanding
M1.04	Describe machinability	3	Understanding
Contents: Metal Cutting: Mechanics of Metal cutting; requirements of tool materials; cutting forces; types of chips; chip thickness ratio; shear angle; types of metal cutting process; orthogonal; oblique and form cutting; Chip Breakers. Cutting speed, feed and depth of cut Tool wear: Types of wear; Tool life; Tool life equations. Machinability: definition; factors affecting machinability; machinability index			
CO2	Illustrate types of cutting tool materials, importance of tool geometry and use of cutting Fluids		
M 2.01	Demonstrate the aspects of Tool materials	3	Applying
M 2.02	Illustrate cutting Tool Geometry	3	Applying
M 2.03	Explain the aspects of cutting Fluids	3	Understanding
M 2.04	Describe the features of lubricants	1	Understanding
	Series Test - I	1	

Contents:

Tool materials: Types; characteristics; applications; Heat treatment of tool steels; Specification of carbide tips; Types of ceramic coatings.

Cutting Tool Geometry: Single point cutting tool; multipoint cutting tools-drills; reamers; milling cutters.

Cutting Fluids & Lubricants: Introduction; Types of cutting fluids, Fluids and coolants required in turning, drilling, shaping, sawing & broaching; Selection of cutting fluids, methods of application of cutting fluid; Classification of lubricants (solid, liquid, gaseous), Properties and applications of lubricants.

CO3	Demonstrate Bench work and fitting, welding and Forging		
M3.01	Describe bench work and fitting & principle of welding	2	Understanding
M3.02	Demonstrate the various welding processes.	3	Applying
M3.03	Explain principle of forging operations	2	Understanding
M3.04	Identify various forging operations	3	Understanding

Contents:

Fitting and Bench work: Vices and their types - filing - methods of filing - classification of files - hammers & chisels and their classification - marking tools - scribes - punches - V block - angle plate - measuring tools (basics only) - hacksaws - blades - setting of teeth - scrapers - drills and reamers - taps and dies.

Forging: Hot working, cold working - advantages of hot working and cold working - hot working operations - rolling, forging, smith forging, drop forging, upset forging, press forging - roll forging

Welding: Classification; Gas welding techniques; Types of welding flames;

Arc Welding - Principle, Equipment, Applications; Shielded metal arc welding; Submerged arc welding; TIG / MIG welding; Resistance welding - Spot welding, Seam welding, Projection welding; Welding defects; Brazing and soldering: Types, Principles, Applications.

CO4	Describe the types of patterns, moulds, and powder metallurgy		
M 4.01	Explain the types of pattern and pattern materials.	3	Understanding
M 4.02	Explain the preparation of sand mould by using the different types of pattern.	3	Understanding
M 4.03	Explain core sand preparation and types of core.	2	Understanding
M 4.04	Recognize the concept of powder metallurgy.	2	Understanding
	Series Test – II	1	

Contents:

Patterns: Definition - types of pattern - solid piece - split piece - loose piece - match plate – sweep - skeleton - segmental - shell - pattern materials - pattern allowances - casting.

Moulding: Definition - Moulding sand - constituents - types - properties of moulding sand - moulding sand preparation - moulding tools - moulding boxes - types of moulds - green sand mould - drysandmould - loam sand mould - methods of moulding - Moulding machines - Jolting - Squeezing - sand slinger.

Cores: Essential qualities of core - materials - core sand preparation - core binders - core boxes - CO2 process core making - types of core.

Powder Metallurgy: Methods of manufacturing metal powders - atomization - reduction and electrolysis deposition - compacting - sintering - sizing - infiltration - mechanical properties of parts made by powder metallurgy - design rules for the powder metallurgy process.

Text/ Reference:

T/R	Book Title/Author
T1	Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt Acharaya, MediaPromoters, 11th Edition, 2007
R1	Production Technology- H.M.T., Tata McGraw Hill
R2	A Text Book of Production engineering – P.C. Sharma, S.Chand& Co.
R3	Introduction of Basic Manufacturing Processes and Workshop Technology, Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006
R4	Production technology – R.K.Jain,Khanna Publishers.

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/
2	http://www.mit.edu/